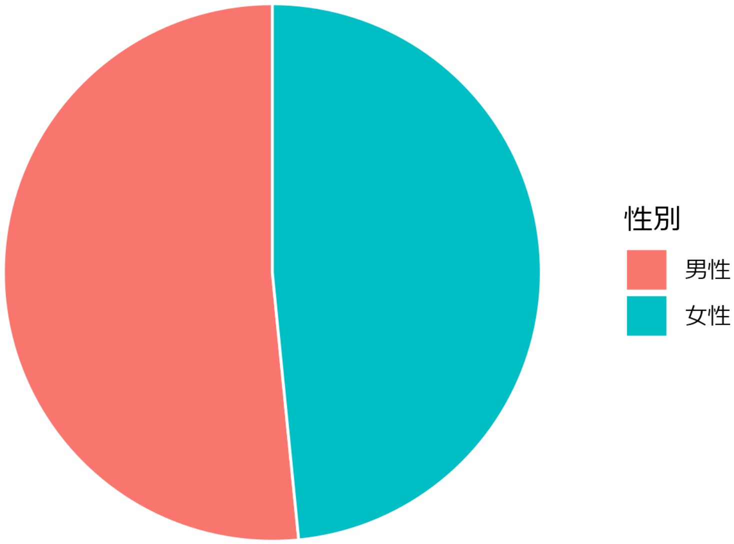
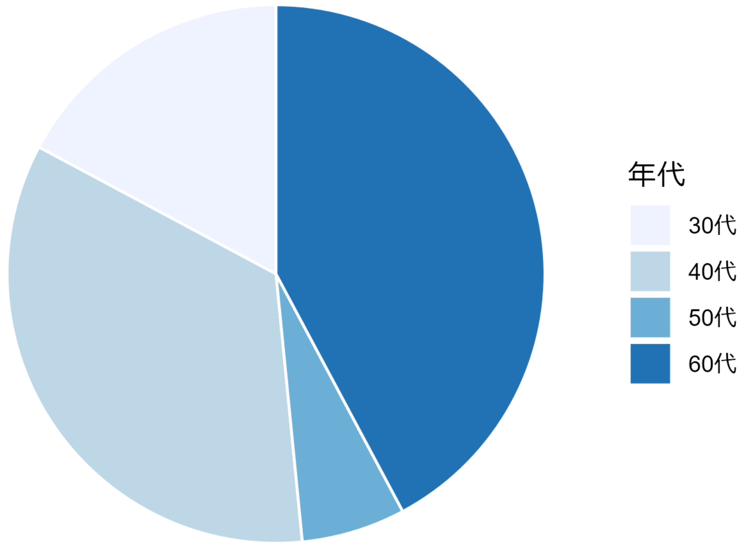
```r
DT::datatable(dfQuest)
```

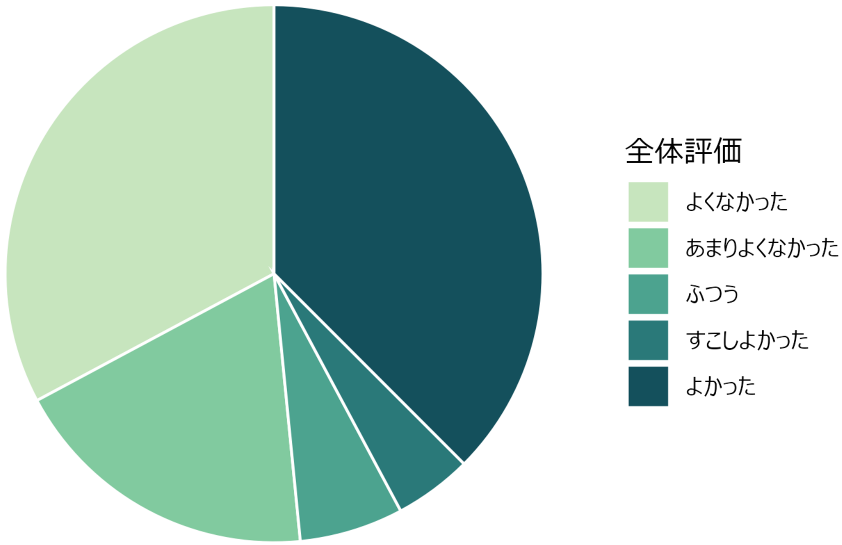
```r
ggplot(
  dfQuest,
  aes(x = "", fill = to_factor(Sex))
) +
  geom_bar(width = 1, color = "white") +
  coord_polar(theta = "y") +
  labs(fill = var_label(dfQuest$Sex)) +
  theme_void()
```




```
ggplot(
  dfQuest,
  aes(x = "", fill = to_factor(Age))
) +
  geom_bar(width = 1, color = "white") +
  coord_polar(theta = "y") +
  scale_fill_brewer(palette = "Blues") +
  labs(fill = var_label(dfQuest$Age)) +
  theme_void()
```




```
ggplot(
  dfQuest,
  aes(x = "", fill = to_factor(Overall))
) +
  geom_bar(width = 1, color = "white") +
  coord_polar(theta = "y") +
  scale_fill_discrete_sequential(palette = "BluGrn") +
  labs(fill = var_label(dfQuest$Overall)) +
  theme_void()
```




```
dfShow <- dfQuest
dfShow$Riyousha_ID <- NULL

dfShow$Sex <- to_factor(dfShow$Sex)
dfShow$Age <- to_factor(dfShow$Age)
dfShow$Overall <- to_factor(dfShow$Overall)

tbl_summary(
  data = dfShow,
  by = Sex
)
```

Fisher p

Characteristic	N = 33[1]	N = 31[1]
	20 (61%)	1 (3.2%)
	9 (27%)	3 (9.7%)
	2 (6.1%)	2 (6.5%)
	0 (0%)	3 (9.7%)
	2 (6.1%)	22 (71%)
30	7 (21%)	4 (13%)
40	18 (55%)	4 (13%)
50	1 (3.0%)	3 (9.7%)
60	7 (21%)	20 (65%)

[1]n (%)

```
dfShow <- dfQuest
dfShow$Riyousha_ID <- NULL

dfShow$Sex <- to_factor(dfShow$Sex)
dfShow$Age <- to_factor(dfShow$Age)
dfShow$Overall <- to_factor(dfShow$Overall)

tbl <- tbl_summary(
  data = dfShow,
  by = Sex
)
tbl <- add_p(tbl)
tbl
```

-
-

Characteristic	N = 33[1]	N = 31[1]	p-value[2]
			<0.001
	20 (61%)	1 (3.2%)	
	9 (27%)	3 (9.7%)	
	2 (6.1%)	2 (6.5%)	
	0 (0%)	3 (9.7%)	
	2 (6.1%)	22 (71%)	
			<0.001
30	7 (21%)	4 (13%)	
40	18 (55%)	4 (13%)	
50	1 (3.0%)	3 (9.7%)	
60	7 (21%)	20 (65%)	

[1]n (%)
[2]Fisher's exact test

: (= y) (= x)

y (/ , YES/NO)

- y
-
- y

```
dfShow <- dfQuest
dfShow$Riyousha_ID <- NULL
dfShow$OverallHigh <- ifelse(as.numeric(dfShow$Overall) >= 4, 1, 0)
dfShow$Overall <- NULL


var_label(dfShow$OverallHigh) <- "   "
val_labels(dfShow$OverallHigh) <- c(
  "  " = 0,
  "  " = 1
```

Characteristic	N = 37[1]	N = 27[1]	**p-value**[2]
			<0.001
	31 (84%)	2 (7.4%)	
	6 (16%)	25 (93%)	
			<0.001
30	7 (19%)	4 (15%)	
40	20 (54%)	2 (7.4%)	
50	3 (8.1%)	1 (3.7%)	
60	7 (19%)	20 (74%)	

[1]n (%)

[2]Pearson's Chi-squared test; Fisher's exact test

```
)

dfShow$OverallHigh <- to_factor(dfShow$OverallHigh)
dfShow$Sex <- to_factor(dfShow$Sex)
dfShow$Age <- to_factor(dfShow$Age)

tbl <- tbl_summary(
  data = dfShow,
  by = OverallHigh
)
tbl <- add_p(tbl)
tbl
```

25/31

2/23

$$= \frac{\frac{25}{31}}{\frac{2}{33}} = 13.3...$$

10

Characteristic	OR	95% CI	p-value
	—	—	
	64.6	14.5, 478	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

$$= \frac{\frac{25}{6}}{\frac{2}{31}} = 96.8...$$

```
fit <- glm(
  OverallHigh ~ Sex,
  family = binomial,
  data = dfShow
)

library(gtsummary)

tbl_regression(
  fit,
  exponentiate = TRUE
)
```

```
fit <- glm(
  OverallHigh ~ Sex + Age,
  family = binomial,
  data = dfShow
)

library(gtsummary)
```

Characteristic	OR	95% CI	p-value
	—	—	
	71.3	12.2, 753	<0.001
30	—	—	
40	0.13	0.01, 1.80	0.15
50	0.08	0.00, 1.72	0.13
60	2.47	0.21, 29.8	0.5

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```r
tbl_regression(
  fit,
  exponentiate = TRUE
)
```

- (71.3)
- 30 60
-

: $(= y)$ $(= x)$

: $(= y)$ $(= x)$

: $(= y)$ $(= x)$